

Fundamental Theorem of Calculus

Date_____ Period_____

For each problem, find $F'(x)$.

1) $F(x) = \int_{-4}^x (t - 1) dt$

2) $F(x) = \int_{-3}^x (t^2 + 2t + 3) dt$

3) $F(x) = \int_{-1}^{x^2} (-2t + 2) dt$

4) $F(x) = \int_4^{3x} (-t^3 + 11t^2 - 39t + 44) dt$

5) $F(x) = \int_2^{x^3} \frac{1}{t^3} dt$

6) $F(x) = \int_x^{x^2} (-2t - 2) dt$

7) $F(x) = \int_x^{x^2} (t^2 - 8t + 11) dt$

8) $F(x) = \int_x^{2x} \frac{2}{t} dt$